

Markov Chain

Thomas Huang

September 5, 2026

Discrete-Time Markov Chains

Overview

We observe the state of a system at discrete times $n = 0, 1, 2, 3, \dots$. Let X_n be the state of the system at time n . Therefore, $\{X_n\}$ is a discrete-time stochastic process. Two extreme cases are:

- $\{X_n\}$ are i.i.d. random variables.
- X_n depends on all past states $X_{n-1}, X_{n-2}, \dots, X_0$.

Markov Property

The Markov property states that given the present state X_n , the future is conditionally independent of the past. Formally,

$$P(X_{n+1} = j \mid X_n = i_n, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = P(X_{n+1} = j \mid X_n = i_n).$$

Thus, the past can only affect the future through the present state.

Definition of a Discrete-Time Markov Chain (DTMC)

A stochastic process $\{X_n, n \geq 0\}$ taking values in a discrete state space S is called a DTMC if:

1. $X_n \in S$ for all $n \geq 0$, with S discrete.
2. $P(X_{n+1} = j \mid X_n = i_n, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = P(X_{n+1} = j \mid X_n = i_n)$ (Markov Property)

If the one-step transition probability does not depend on time n , i.e.,

$$P_{ij}(n) = P(X_{n+1} = j \mid X_n = i) = P(X_1 = j \mid X_0 = i) = P_{ij},$$

we call the Markov chain *time-homogeneous*. In that case, $P = [P_{ij}]$ is the transition probability matrix of going from state i to j , with

- $0 \leq P_{ij} \leq 1$ for all P_{ij}
- $\sum_{j \in S} P_{ij} = 1$ for each i . (All row-sum = 1)

Such matrix is called stochastic matrix.

Initial Distribution

A DTMC is characterized by:

- The transition probability matrix P .
- The initial distribution $\mathbf{a} = [a_1, a_2, \dots, a_m]$, where $a_j = P(X_0 = j)$.

Hence, jointly they specify the distribution of the entire chain.

Characterization

$[P]$ is not sufficient to characterize a DTMC because P_{ij} is not finite dimension joint distribution.

$$\begin{aligned} P(X_1 = 5, X_0 = 1) &= P(X_1 = 5 | X_0 = 1)P(X_0 = 1) \\ &= P_{15}P(X_0 = 1) \\ &= P_{15}a_1 \end{aligned}$$

A DTMC is uniquely characterize by $[P]$ and its initial distribution $\mathbf{a} = [a_1, a_2, \dots, a_j, \dots, a_m]$ where $a_j = P(X_0 = j)$

Examples

Example 1: A Two-State Machine

A machine can be in one of two states: “up” or “down”. If it is up on day n , it remains up on day $n + 1$ with probability P_u , otherwise it goes down with probability $1 - P_u$. If it is down on day n , it stays down on day $n + 1$ with probability P_d , otherwise it goes up with probability $1 - P_d$. Denote X_n as the state of the machine on day n (“up” or “down”).

Let X_n = state of the machine on day n .

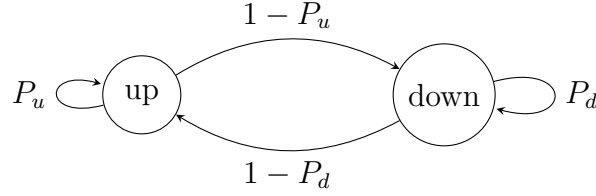
This is a Markov chain with state space $S = \{\text{up}, \text{down}\}$ and transition probabilities:

$$\begin{aligned} P(X_{n+1} = \text{up} \mid X_n = \text{up}) &= P_u \\ P(X_{n+1} = \text{down} \mid X_n = \text{up}) &= 1 - P_u \\ P(X_{n+1} = \text{up} \mid X_n = \text{down}) &= 1 - P_d \\ P(X_{n+1} = \text{down} \mid X_n = \text{down}) &= P_d. \end{aligned}$$

Then the transition probability matrix $[P]$ is

$$[P] = \begin{array}{c|cc} & \text{to up} & \text{to down} \\ \hline \text{from up} & P_u & 1 - P_u \\ \text{from down} & 1 - P_d & P_d \end{array}$$

Transition Diagram



- nodes $\Rightarrow$ states (each node represents a state)
- arcs $\Rightarrow$ transition

Example 2: Two Independent Machines

Consider two identical machines operating independently. Each machine transitions between “up” and “down” states as above.

Let X_n = number of machines that are “up” at time n .
Then $S = \{0, 1, 2\}$.

For each machine:

$$\begin{aligned} up &\Rightarrow up \text{ with prob. } P_u \\ down &\Rightarrow down \text{ with prob. } P_d \end{aligned}$$

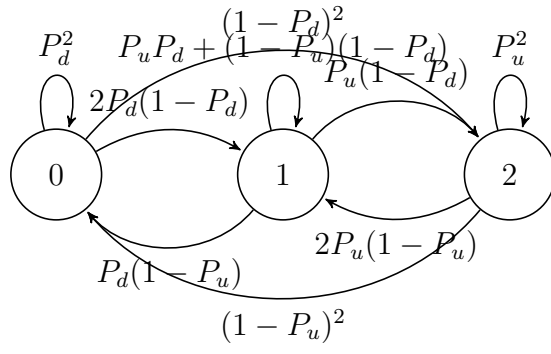
The transition probabilities can be computed based on the independence of the individual machines.

$$\begin{aligned} P(X_{n+1} = 0 \mid X_n = 0) &= P(\text{Both } M_{1,2} \text{ down} \mid \text{Both } M_{1,2} \text{ down}) = P_d^2 \\ P(X_{n+1} = 1 \mid X_n = 0) &= P(\text{Either } M_{1,2} \text{ up} \mid \text{Both } M_{1,2} \text{ down}) = 2P_d(1 - P_d) \\ P(X_{n+1} = 0 \mid X_n = 1) &= P(\text{Both } M_{1,2} \text{ down} \mid \text{Either } M_{1,2} \text{ up}) = P_d(1 - P_u) \\ P(X_{n+1} = 1 \mid X_n = 1) &= P(\text{Either } M_{1,2} \text{ up} \mid \text{Either } M_{1,2} \text{ up}) = P_u P_d + (1 - P_u)(1 - P_d) \\ P(X_{n+1} = 2 \mid X_n = 0) &= P(\text{Both } M_{1,2} \text{ up} \mid \text{Both } M_{1,2} \text{ down}) = (1 - P_d)^2 \\ P(X_{n+1} = 0 \mid X_n = 2) &= P(\text{Both } M_{1,2} \text{ down} \mid \text{Both } M_{1,2} \text{ up}) = (1 - P_u)^2 \\ P(X_{n+1} = 2 \mid X_n = 1) &= P(\text{Both } M_{1,2} \text{ up} \mid \text{Either } M_{1,2} \text{ up}) = P_u(1 - P_d) \\ P(X_{n+1} = 1 \mid X_n = 2) &= P(\text{Either } M_{1,2} \text{ up} \mid \text{Both } M_{1,2} \text{ up}) = 2P_u(1 - P_u) \\ P(X_{n+1} = 2 \mid X_n = 2) &= P(\text{Both } M_{1,2} \text{ up} \mid \text{Both } M_{1,2} \text{ up}) = P_u^2 \end{aligned}$$

Then the transition probability matrix $[P]$ is

$$[P] = \begin{array}{c|ccc} & \text{to 0 up} & \text{to 1 up} & \text{to 2 up} \\ \hline \text{from 0 up} & P_d^2 & 2P_d(1 - P_d) & (1 - P_d)^2 \\ \text{from 1 up} & P_d(1 - P_u) & P_u P_d + (1 - P_u)(1 - P_d) & P_u(1 - P_d) \\ \text{from 2 up} & (1 - P_u)^2 & 2P_u(1 - P_u) & P_u^2 \end{array}$$

Transition Diagram



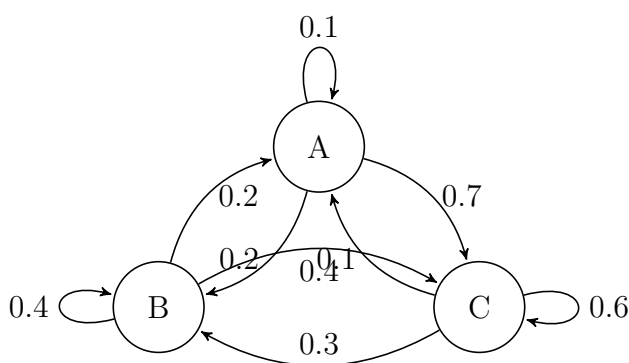
Example 3: Brand Switching

A customer chooses one of three brands A, B, C of beer every day, and the choice on day $n + 1$ depends only on the brand chosen on day n . State space is $S = \{A, B, C\}$. A typical transition probability matrix might be:

$$P = \begin{pmatrix} 0.1 & 0.2 & 0.7 \\ 0.2 & 0.4 & 0.4 \\ 0.1 & 0.3 & 0.6 \end{pmatrix}.$$

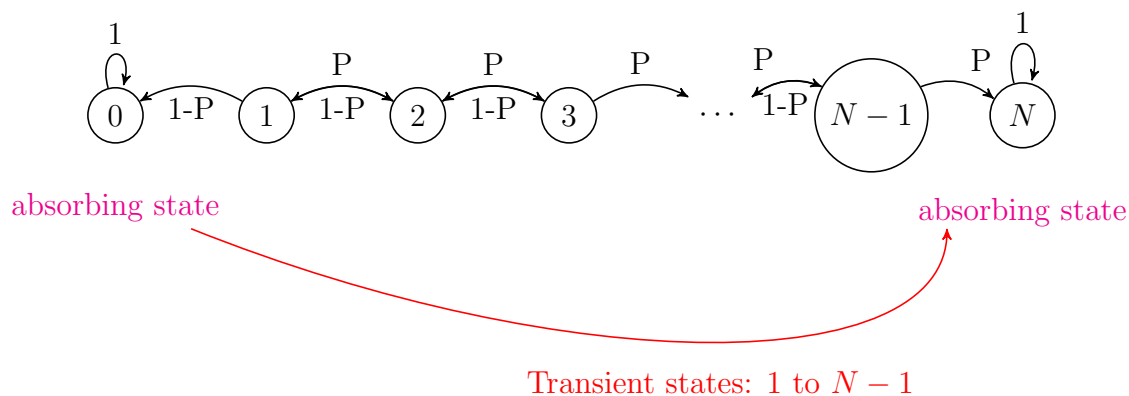
The entry in row i , column j represents $P(X_{n+1} = j \mid X_n = i)$.

Transition Diagram



Example 4: Gambler's Ruin

Person A and B share a total of $\$N$. Each bet $\$1$ on the flip of a coin. If the coin is heads (probability p), A wins $\$1$ from B ; otherwise A loses $\$1$ to B . Let X_n be the amount of money A has after the n -th bet. Then X_n takes values in $\{0, 1, \dots, N\}$, and we have a Markov chain with absorbing states 0 and N (since once A has 0 or N , the game ends).



Transient Behavior(Marginal Distribution)

The n -step transition probability is defined by

$$P_{ij}^{(n)} = P(X_n = j \mid X_0 = i).$$

The Chapman-Kolmogorov equation states:

$$P_{ij}^{(n)} = \sum_{r \in S} P_{ir}^{(k)} P_{rj}^{(n-k)}, \quad 0 \leq k \leq n.$$

For a time-homogeneous DTMC,

$$P^{(n)} = P^n,$$

where $P^{(n)} = [P_{ij}^{(n)}]$ is the n -step transition matrix. The distribution of X_n is then

$$P(X_n = j) = \sum_{i \in S} P(X_n = j \mid X_0 = i) P(X_0 = i) = \sum_{i \in S} P_{ij}^{(n)} a_i,$$

or in vector form,

$$\mathbf{a}^{(n)} = \mathbf{a} P^{(n)} = \mathbf{a} P^n.$$

Example

Let $\{X_n\}$ be a DTMC. Let $S = \{1, 2, 3, 4\}$ and

$$[P] = \begin{bmatrix} 0.1 & 0.2 & 0.3 & 0.4 \\ 0.2 & 0.2 & 0.3 & 0.3 \\ 0.5 & 0 & 0.5 & 0 \\ 0.6 & 0.2 & 0.1 & 0.1 \end{bmatrix}$$

and initial distribution

$$\mathbf{a} = \begin{bmatrix} 0.25 \\ 0.25 \\ 0.25 \\ 0.25 \end{bmatrix}$$

Find $P(X_3 = 4, X_2 = 1, X_1 = 3, X_0 = 1)$, $P(X_3 = 4, X_2 = 1, X_1 = 3)$, $a^{(n)}$

(1)

$$\begin{aligned} P(X_3 = 4, X_2 = 1, X_1 = 3, X_0 = 1) &= P(X_3 = 4 \mid X_2 = 1, X_1 = 3, X_0 = 1) \\ &\quad P(X_2 = 1 \mid X_1 = 3, X_0 = 1) \\ &\quad P(X_1 = 3 \mid X_0 = 1) P(X_0 = 1) \\ &= P(X_3 = 4 \mid X_2 = 1) P(X_2 = 1 \mid X_1 = 3) \\ &\quad P(X_1 = 3 \mid X_0 = 1) P(X_0 = 1) \\ &= p_{14} p_{13} p_{13} a_1 \\ &= 0.4 \times 0.5 \times 0.3 \times 0.25 \\ &= 0.015 \end{aligned}$$

(2)

$$\begin{aligned}
P(X_3 = 4, X_2 = 1, X_1 = 3) &= \sum_{i=1}^4 P(X_3 = 4, X_2 = 1, X_1 = 3, X_0 = i) \\
&= \sum_{i=1}^4 (P(X_3 = 4 \mid X_2 = 1, X_1 = 3, X_0 = i) \\
&\quad P(X_2 = 1 \mid X_1 = 3, X_0 = i) \\
&\quad P(X_1 = 3 \mid X_0 = i) P(X_0 = i)) \\
&= P(X_3 = 4 \mid X_2 = 1, X_1 = 3, X_0 = i) \\
&\quad P(X_2 = 1 \mid X_1 = 3, X_0 = i) \\
&\quad \sum_{i=1}^4 (P(X_1 = 3 \mid X_0 = i) P(X_0 = i)) \\
&= p_{31} p_{14} \left(\sum_{i=1}^4 a_i p_{i3} \right) \\
&= 0.5 \times 0.4 \times [0.25 \times (0.3 + 0.3 + 0.5 + 0.1)] \\
&= 0.06
\end{aligned}$$

(3)

$$\begin{aligned}
a^{(4)} &= p^{(4)} a^{(0)} = p^4 a, \\
a^{(1)} &= \begin{pmatrix} 0.341 \\ 0.135 \\ 0.325 \\ 0.198 \end{pmatrix}
\end{aligned}$$

Limiting Behavior of a DTMC

$$\begin{aligned}
\lim_{n \rightarrow \infty} a_j^{(n)} &= \lim_{n \rightarrow \infty} P(X_n = j) \\
a^{(n)} &= P^n a \\
\lim_{n \rightarrow \infty} a^{(n)} &= \lim_{n \rightarrow \infty} P^n a \\
\lim_{n \rightarrow \infty} a^{(n)} &= a \lim_{n \rightarrow \infty} P^n
\end{aligned}$$

Example: Brand Switching

A customer buys beer every day among 3 different brands A, B, C . The choice on day $n+1$ depends only on the choice on day n . Let X_n be the brand choice on day n . Suppose

$$X_n \in \{A, B, C\}, \quad P = \begin{bmatrix} 0.1 & 0.2 & 0.7 \\ 0.2 & 0.4 & 0.4 \\ 0.1 & 0.3 & 0.6 \end{bmatrix}$$

Suppose the initial distribution is

$$a = [0.2 \quad 0.3 \quad 0.5].$$

The question is: What is the long-run market share of a particular brand?

$$a^{(1)} = \begin{bmatrix} 0.13 \\ 0.31 \\ 0.56 \end{bmatrix} \quad \text{As } n \text{ gets larger, } P^n = \begin{bmatrix} 0.132 & 0.319 & 0.549 \\ 0.132 & 0.319 & 0.549 \\ 0.132 & 0.319 & 0.549 \end{bmatrix}.$$

Hence, the limit distribution exists (the rows become identical). Therefore, no matter what a is,

$$\lim_{n \rightarrow \infty} a^{(n)} = [0.132 \quad 0.319 \quad 0.549].$$

(The limiting distribution is unaffected by the choice of a .)

What if

$$[P] = \begin{bmatrix} 0 & 1 & 0 \\ 0.5 & 0 & 0.5 \\ 0 & 1 & 0 \end{bmatrix}$$

When n gets larger then

$$P^{(2k)} = \begin{bmatrix} 0.5 & 0 & 0.5 \\ 0 & 1 & 0 \\ 0.5 & 0 & 0.5 \end{bmatrix} \quad P^{(2k+1)} = \begin{bmatrix} 0 & 1 & 0 \\ 0.5 & 0 & 0.5 \\ 0 & 1 & 0 \end{bmatrix}$$

Their corresponded limiting distribution is

$$\begin{aligned} a^{(2k)} &= a \cdot P^{(2k)} \\ &= \begin{bmatrix} 0.35 \\ 0.3 \\ 0.35 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} a^{(2k+1)} &= a \cdot P^{(2k+1)} \\ &= \begin{bmatrix} 0.15 \\ 0.7 \\ 0.15 \end{bmatrix} \end{aligned}$$

limiting distribution does not exist also the distribution depends on the choices of $\mathbf{a}$.

So, under what condition does limiting distribution exist?

- periodicity

If it exists limiting distribution, when is it dependent of the choice of the initial distribution $\mathbf{a}$?

- irreducibility

Section 6.2.1 Classification of States (Irreducibility, periodicity)

Accessible

A state j is said to be *accessible* from state i if there exists an n such that

$$P_{ij}^{(n)} = P(X_n = j \mid X_0 = i) > 0.$$

Write $i \rightarrow j$ if i is accessible to j . Clearly $i \rightarrow i$ for any state i .

Communicate

A state i is said to *communicate* with j (written $i \leftrightarrow j$) if $i \rightarrow j$ and $j \rightarrow i$.

Theorem

If $i \leftrightarrow j$ and $j \leftrightarrow k$, then $i \leftrightarrow k$.

Communicating class

A set $C \subseteq S$ is said to be a *communicating class* if for any $i, j \in C$, $i \leftrightarrow j$ holds.

Closeure

A communicating class C is said to be *closed* if whenever $i \in C$ and $i \rightarrow j$, then $j \in C$. Otherwise, C is *open*.

If $X_n \in C$ for some n , then $X_m \in C$ for all $m \geq n$.

Definition

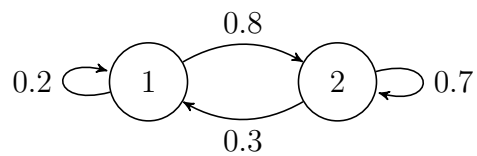
A DTMC is said to be *irreducible* if all its states belong to a single closed communicating class. Otherwise it is *reducible*.

Example

Example 1:

$$S = \{1, 2\}$$

$$P = \begin{pmatrix} 0.2 & 0.8 \\ 0.3 & 0.7 \end{pmatrix}$$



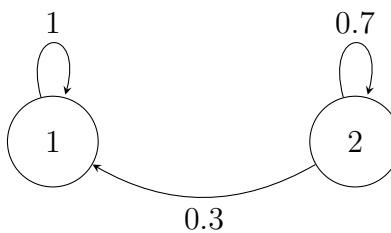
Relationship:

- $1 \leftarrow 2$
- $1 \rightarrow 2$
 - Since $1 \leftarrow 2$ and $1 \rightarrow 2$ So, $1 \leftrightarrow 2$
- $1 \leftrightarrow 1$
- $2 \leftrightarrow 2$

So this chain is irreducible since it consists only one communication class

Example 2:

$$P = \begin{pmatrix} 1 & 0 \\ 0.3 & 0.7 \end{pmatrix} \quad S = \{1, 2\}$$



Relationship:

- $1 \leftrightarrow 1$
 - 1 is a closed communication class
- $2 \rightarrow 1$
- $2 \leftrightarrow 2$

- 2 is a opened communication class

So this chain is reducible since it consists two communication classes

Example 3:

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$



Relationship:

- $1 \leftrightarrow 1$
 - 1 is a closed communication class
- $2 \leftrightarrow 2$
 - 2 is a closed communication class

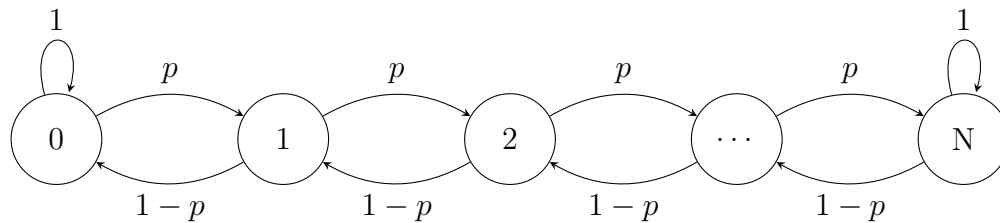
So this chain is reducible since it consists two communication classes

Example 3: Random Walk on $0, 1, \dots, N$

p = probability of stepping right

$(1 - p)$ = probability of stepping left

Random walk states: $\{0, 1, \dots, N\}$, Transitions: $P(i \rightarrow i + 1) = p$, $P(i \rightarrow i - 1) = 1 - p$



Relationship:

- $0 \leftrightarrow 0$
 - $\{0\}$ is a closed communication class
- $N \leftrightarrow N$

- $\{N\}$ is a closed communication class
- $0 \leftarrow 1$
- $N - 1 \rightarrow N$
- $i \leftrightarrow i + 1$ for $i = 1, 2, \dots, N - 1$
- $\{1, 2, \dots, N - 1\}$ is a opened communication class

So this chain is reducible since it consists three communication classes

Periodicity

A state i is said to be *periodic* with period d if d is the largest integer such that

$$P_{ii}^{(n)} = P(X_n = i | X_0 = i) > 0 \Rightarrow \text{is an integer multiple of } d$$

If $d = 1$, i is called *aperiodic*.

Ex) The set of integers n for which $p_{ii}^{(n)} > 0$ is given by:

$$\begin{aligned} \{0, 3, 7, 9, 11, \dots\} &\Rightarrow \text{gcd} = 1 \text{ (aperiodic),} \\ \{0, 3, 6, 9, 12, 15, \dots\} &\Rightarrow \text{gcd} = 3 \text{ (periodic),} \\ \{0, 1, 3, 3, 6, 9, 12, 15, \dots\} &\Rightarrow \text{gcd} = 1 \text{ (aperiodic).} \end{aligned}$$

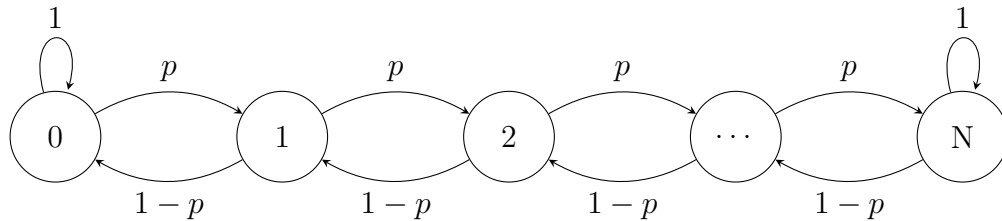
Theorem: If $i \leftrightarrow j$, then i and j have the same period. All states in a communicating class share the same period.

Ex) In a chain with transition probability p , suppose

$$P_{00} = 1 > 0 \Rightarrow d = 1 \text{ (aperiodic),}$$

and if $P_{nn} = 1 > 0$, then $d = 1$ again (aperiodic). However, if from any state one needs $2n$ steps to return to state 1, then $d = 2$ (periodic).

Ex)



- For $\{0\}$ Class: $P_{00} = 1 > 0$, so $d = 1$. Therefore, aperiodic.
- For $\{1, 2, \dots, N - 1\}$ Class: Pick a random state. For example, 1, then the steps to come back to state 1 from any state is $2n$, so $d = 2$. Therefore, periodic.
- For $\{N\}$ Class: $P_{NN} = 1 > 0$, so $d = 1$. Therefore, aperiodic.

Recurrence and Transience

When thinking about the long-run behaviour of Markov chains, it's useful to classify two different types of states: *recurrent* states and *transient* states.

Recurrent states	Transient states
If we ever visit i , then we keep returning to i again and again	We might visit i a few times, but eventually we leave i and never come back
Starting from i , the expected number of visits to i is infinite	Starting from i , the expected number of visits to i is finite
Starting from i , the number of visits to i is certain to be infinite	Starting from i , the number of visits to i is certain to be finite
The return probability f_i equals 1	The return probability f_i is strictly less than 1

Table 1: Comparison of recurrent and transient states in Markov chains.

Definition of First Passage Time

Fix a state i . Define the first passage time:

$$T_i = \min\{n > 0 : X_n = i\}$$

T_i is an integer-valued random variable.

Let $f_i = P(T_i < \infty | X_0 = i)$ be the probability that state i will be revisited by the MC.

Recurrent and Transient States

Definition

A state i is called:

- recurrent if $f_i = 1$
- transient if $f_i < 1$

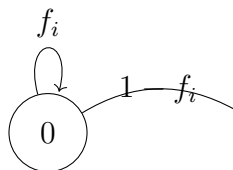
Intuitive Interpretation

Let N = Total times state i is visited by the MC

- For recurrent i : $E[N|X_0 = i] = \infty$



- For transient i : $E[N|X_0 = i] = \frac{1}{1-f_i}$



- $P(N = k|X_0 = i) = f_i^{k-1}(1 - f_i)$

Necessary and Sufficient Conditions

Theorem: A state i is:

- recurrent if and only if $\sum_{n=0}^{\infty} P_{ii}^{(n)} = \infty$
- transient if and only if $\sum_{n=0}^{\infty} P_{ii}^{(n)} < \infty$

Proof Sketch. Let $Y_n = \begin{cases} 1 & \text{if } X_n = i \\ 0 & \text{if } X_n \neq i \end{cases}$

Then:

$$E[N|X_0 = i] = E\left[\sum_{n=0}^{\infty} Y_n | X_0 = i\right] = \sum_{n=0}^{\infty} E[Y_n | X_0 = i] = \sum_{n=0}^{\infty} P_{ii}^{(n)}$$

□

Class Properties

1. If state i is recurrent and $i \leftrightarrow j$, then j is also recurrent
2. If state i is transient and $i \leftrightarrow j$, then j is also transient
3. An irreducible DTMC is recurrent/transient if all of its states are recurrent/transient

Finite Communicating Classes

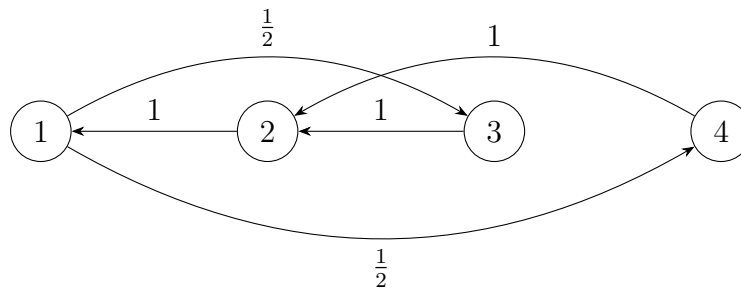
1. Let C be a finite closed communicating class. All states in C are recurrent.
2. Let C be a finite open communicating class. All states in C are transient.
3. If S is finite, all communicating classes are finite (either open or closed).

Example

(1) Given a transition matrix:

$$[P] = \begin{bmatrix} 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

We can create a transition diagram:

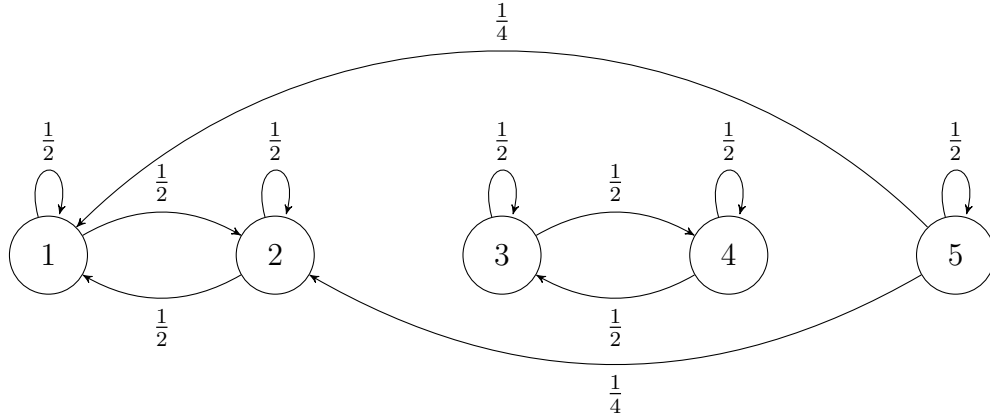


- $\{1, 2, 3, 4\}$ forms a single communicating class
- The chain is irreducible
- The class is closed
- Therefore, all states are recurrent

(2) Given a transition matrix:

$$[P] = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{4} & 0 & 0 & \frac{1}{2} \end{bmatrix}$$

We can create a transition diagram:



- $\{1, 2\}$ closed recurrent communicating class
- $\{3, 4\}$ closed recurrent communicating class
- $\{5\}$ open transient communicating class

Limiting Behavior of Irreducible DTMCs

Suppose the state space S is finite.

Theorem 1

For finite irreducible DTMCs, there exists a unique stationary distribution vector

$$\pi = [\pi_1, \pi_2, \dots, \pi_m]$$

, satisfying:

- $\pi P = \pi$,
- $\sum_{j=1}^m \pi_j = 1$,
- Each entry of $\pi_j > 0$.

Theorem 2 (Ergodic Theorem)

For finite irreducible aperiodic DTMCs:

- The limiting probability exists:

$$P_{ij}^{(n)} \rightarrow \pi_j, \quad n \rightarrow \infty.$$

- In the aperiodic case: π is the limiting distribution.
- The stationary distribution satisfies:

- Long-run fraction of time spent in state j :

$$E[\text{time in } j] = \pi_j.$$

- if $a = \pi$ then π is called the steady state distribution.

Theorem 3

For finite irreducible periodic DTMCs:

- The limiting probability $P_{ij}^{(n)}$ does not exist.
- However, a stationary distribution still exists and satisfies:

$$\pi P = \pi.$$

Examples

Example 1: Brand Switching

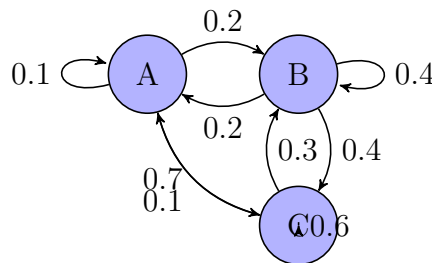
Consider a DTMC with three states (A, B, C) , transition matrix:

$$P = \begin{bmatrix} 0.1 & 0.2 & 0.7 \\ 0.2 & 0.4 & 0.4 \\ 0.1 & 0.3 & 0.6 \end{bmatrix}.$$

Initial distribution:

$$a^{(0)} = [a_A, a_B, a_C] = [0.2, 0.3, 0.5].$$

Transition Diagram:



Long-Run Behavior: As $n \rightarrow \infty$, the long-run market share stabilizes at:

$$\pi = [\pi_A, \pi_B, \pi_C] = [0.132, 0.319, 0.549].$$

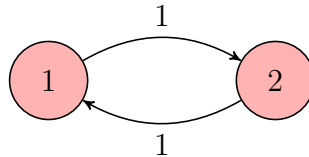
Example 2: Periodic DTMC

Consider the following transition matrix:

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

This chain is periodic with period $d = 2$.

Transition Diagram:



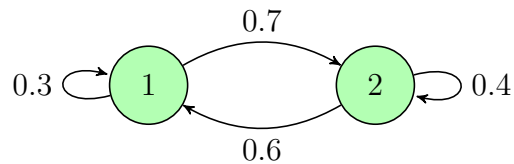
In this case: The limiting distribution does not exist due to periodicity.

Example 3: Aperiodic DTMC

Consider the following transition matrix:

$$P = \begin{bmatrix} 0.3 & 0.7 \\ 0.6 & 0.4 \end{bmatrix}.$$

This chain is irreducible and aperiodic. Transition Diagram:



In this case: The limiting distribution exists and is given by:

$$\pi = [\pi_1, \pi_2] = [0.462, 0.538].$$